

Build Job Log: Selective Laser Melting of ER70 Steel Marine Propeller

Job ID: SLM-2023-087

Product: Marine Propeller (ER70 Steel)

Build Date: October 15, 2023

Operator: J. Doe

Machine: SLM 280 HL

Post-Processing Equipment: CNC Mill VM-2

Job Overview

This log documents the complete manufacturing process for a single ER70 steel marine propeller produced via Selective Laser Melting (SLM) with a 1 mm machining allowance. The process involved three primary phases: powder feedstock production via gas atomization, the SLM build itself, and final finish machining. All operational parameters, material consumptions, and energy usages are recorded as per standard procedure.

Key specifications: - Target component weight: 0.204 kg - Build platform: 280 x 280 mm - Layer thickness: 30 µm - Laser power: 400 W - Scan speed: 800 mm/s - Support structures: Custom lattice design

Powder Production Phase – Gas Atomization

Date: October 10, 2023

Equipment: Argon Gas Atomizer Unit GA-200

Operator: A. Smith

The ER70 steel powder was produced in-house using gas atomization. The process commenced at 08:00 and concluded at 12:30. Material and utility consumptions were monitored via integrated sensors.

Process Parameters: - Melt temperature: 1650°C - Argon pressure: 4.5 bar - Water cooling flow rate: 12 L/min - Atomization duration: 4.5 hours

Material and Energy Consumption:

Item	Quantity	Unit	Notes
Steel billet input	0.414	kg	ER70 grade, certified
Electricity consumption	0.828	kWh	Mainly for induction melting and gas compression
Argon gas usage	2.58	kg	High-purity, 99.998%
Cooling water	0.116	kg	Closed-loop system, make-up water only
Powder output	0.352	kg	Sieved to 15-45 µm, yield ~85%

Sensor Readings (Averaged): - Melt chamber temperature: 1645-1655°C - Argon flow rate: 55 L/min - Power draw: 0.184 kW (average over 4.5 h) - Cooling water temperature out: 35°C

No alarms or deviations recorded. Powder quality met specification with spherical morphology >95%.

SLM Build Phase

Date: October 15, 2023

Equipment: SLM 280 HL

Operator: J. Doe

Build Start: 06:00

Build End: 12:33

Total Build Time: 6.55 hours

The build process was executed with standard parameters for ER70 steel. The job file “Propeller_B1.slm” was loaded, and the build chamber was purged and prepared.

Build Parameters: - Layer count: 1100 - Recoater speed: 150 mm/s - Chamber oxygen level: <0.1% - Build plate temperature: 200°C

Operational Log:

Time	Event / Parameter	Value	Unit	Notes
06:00	Build start	-	-	Chamber sealed, purge initiated
06:15	Oxygen level	0.08	%	Within spec
06:20	Powder deposition	-	-	First layer completed
06:30	Laser activation	-	-	Melting commenced
06:30-12:30	Average power draw	1.75	kW	Consistent throughout build
12:33	Build complete	-	-	Laser off, cool-down initiated

Material and Utility Consumption:

Item	Quantity	Unit	Notes
ER70 steel powder	0.352	kg	Including support structures
Electricity consumption	11.49	kWh	Total for build cycle
Compressed air	3.56	m³	For powder handling and cleaning
Argon gas	1.2	m³	For atmosphere control

Sensor Data (Averaged): - Laser power: 400 W - Chamber temperature: 195-205°C - Argon flow: 3.0 L/min - Compressed air pressure: 6 bar

No system alarms. Build completed successfully with 100% density confirmed via in-process monitoring.

Finish Machining Phase

Date: October 16, 2023

Equipment: CNC Mill VM-2

Operator: R. Brown

Start Time: 08:00

End Time: 12:08

Total Machining Time: 4.133 hours

The propeller was removed from the build plate and transferred to the CNC mill for finish machining to achieve final dimensions and surface finish. A 1 mm machining allowance was removed from all functional surfaces.

Machining Parameters: - Spindle speed: 2500 RPM - Feed rate: 200 mm/min - Depth of cut: 0.5 mm (two passes) - Tool: Carbide end mill, 10 mm diameter

Operational Log:

Time	Event / Parameter	Value	Unit	Notes
08:00	Part setup	-	-	Fixturing completed
08:15	Coolant system on	-	-	Cutting fluid circulated
08:20	Machining start	-	-	First tool path
08:20-12:08	Average power draw	0.687	kW	Steady operation
12:08	Machining complete	-	-	Part unloaded

Material and Energy Consumption:

Item	Quantity	Unit	Notes
Electricity consumption	2.84	kWh	For spindle, axes, and auxiliaries
Cutting fluid	1.94	kg	Water-soluble oil, 5% concentration
Finished propeller	0.204	kg	1 unit, meets drawing specs
Machined chips (waste)	0.116	kg	ER70 steel, collected for recycling

Sensor Readings: - Spindle load: 65-75% - Coolant temperature: 22°C - Tool wear: Within limits, no tool change required

No operational issues. Surface finish measured at Ra 1.6 µm, within tolerance.

Summary of Material and Energy Flows

Consolidated Data for Current Build:

Phase	Material / Utility	Quantity	Unit
Powder Production	Steel billet	0.414	kg
Powder Production	Electricity	0.828	kWh
Powder Production	Argon	2.58	kg
Powder Production	Water	0.116	kg
SLM Build	Steel powder	0.352	kg
SLM Build	Electricity	11.49	kWh
SLM Build	Compressed air	3.56	m³
SLM Build	Argon	1.2	m³
Finish Machining	Electricity	2.84	kWh
Finish Machining	Cutting fluid	1.94	kg
Output	Marine propeller	0.204	kg
Waste	Machined chips	0.116	kg

Contextual Performance Metrics: - Gas atomization electricity intensity: 2 kWh/kg powder (calculated from process data) - SLM build efficiency: 0.031 kg/h (based on 6.55 h build time) - Machining material removal rate: 0.028 kg/h

Equipment Status and Maintenance Notes

SLM 280 HL Post-Build Check: - Recoater blade: No wear observed - Optics: Clean, no contamination - Filter unit: 15% life remaining - Build plate: Minimal distortion, within tolerance

CNC Mill VM-2 Post-Operation: - Tool holder: Secure - Coolant system: No leaks - Axis alignment: Within spec

No unscheduled maintenance required. All equipment is ready for next job.

Comparative Data (Historical Reference)

Note: The following data is for contextual comparison and is not part of the current build record.

- **Previous similar build (Job SLM-2023-045):**
SLM electricity: ~12.1 kWh, Machining electricity: ~3.0 kWh
Powder usage: ~0.360 kg, Yield: ~0.200 kg
- **Industry benchmark for SLM of steel components:**
Typical electricity: 10-15 kWh per build, Powder efficiency: 80-90%

The current build showed improved powder utilization and slightly lower energy consumption compared to previous jobs, likely due to optimized support structures and laser parameters.

Sign-off

Operators:

J. Doe (SLM), R. Brown (Machining)

Quality Check:

Part dimensions and surface finish verified. No defects.

Log Closed: October 16, 2023, 14:00